Slew Rate Impact on Two Metrics
Slew Rate and Shot Rate Deep Dive — Why $\dot{\theta}_{max}$ Shows No Effect on Prediction Error
Prediction error: zero sensitivity (KF ≠ mirror)
Targeting gap: collapses at critical threshold
Targeting gap P95 $\|p_l - \hat{r}\|$
Prediction error P95 $\|\hat{r} - r_{true}\|$
20 µm req.
Flat: KF prediction independent of mirror
Error [µm]
Mirror slew rate $\ddot{\theta}_{max}$ [rad/s²]
Available Steering Budget During τ_l
Below critical: mirror cannot reach $\hat{r}$
Available: $\delta = \frac{1}{2}\ddot{\theta}\,\tau_l^2$
P95 required ≈ 117 µrad
Slew-limited zone
Nominal θ=1000 rad/s² (38× margin)
Critical $\ddot{\theta}^* = 26$ rad/s²
Available steering δ [µrad] (log)
$\ddot{\theta}_{max}$ [rad/s²]
Shot Rate ($f_s = 5$ Hz) vs Latency Budget
Inter-shot gap is 200 ms vs 3 ms latency → 66× more time
During latency ($\tau_l = 3$ ms)
Inter-shot gap ($f_s = 5$ Hz → 200 ms)
P95 required ≈ 117 µrad
Inter-shot gap >300×
larger than latency
→ shot rate irrelevant
Available δ [µrad]
$\ddot{\theta}_{max}$ [rad/s²]
Critical Shot Rate Where Inter-Shot
Repositioning Starts Limiting Accuracy
Critical $f_s^*(\ddot{\theta})$: inter-shot starts to bind
Nominal $f_s = 5$ Hz
1 kHz (physically extreme)
Safe zone (current)
At 5 Hz, shot rate
only matters for
θ < 0.01 rad/s²
Critical shot rate f_s^* [Hz]
$\ddot{\theta}_{max}$ [rad/s²]